Maria Magdalena DICU

ASSOCIATE PROFESSOR, PhD IN MATERIALS SCIENCE AND ENGINEERING



CONTACT DETAILS

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ACADEMIC PROFILE

ORCID 0000-0003-4079-2703
Nationality: Romanian

AREAS OF EXPERTISE

Materials Science and Engineering · Biomaterials · Composite Materials · Coatings and Thin Films · Electrochemistry and Materials Characterisation · Materials Engineering

LANGUAGES

French: C2 · English: B1–B2

DIGITAL SKILLS

Microsoft Office · Sigma Plot · Origin · CMPR Logic · EndNote · Software for analytical and characterisation instruments

PROFESSIONAL PROFILE

University lecturer and researcher in materials science and engineering. Her work brings together advanced materials research, project coordination, student education and doctoral mentoring.

CURRENT POSITION

Associate Professor since February 2022, Faculty of Mechanics and Technology, Manufacturing and Industrial Management Department, Pitești University Centre – POLITEHNICA Bucharest.

PROFESSIONAL EXPERIENCE

Feb. 2022–present | Associate Professor
National University of Science and Technology POLITEHNICA Bucharest, Pitești University Centre, Faculty of Mechanics and Technology, Manufacturing and Industrial Management Department

- Teaching courses in Materials Science and Engineering, Materials Science, Materials Technology, Manufacturing Technologies for Semi-Finished Products, Biomaterials, Composite Materials, Materials Characterisation, Phase Transformations and Materials Microstructure, and Applied Thermodynamics in Metallurgy.
- Supervising undergraduate projects in Industrial Economic Engineering and Machine Building Technology, as well as master's dissertations in Materials Science and Technology (STM).
- Director of the STM master's programme; member of doctoral programme and research proposal committees in Materials Science; mentor to doctoral students in Materials Engineering; supervisor of student research papers.

2021–present | Director of the French-Romanian Master's Programme in Materials Science and Technology (STM)
POLITEHNICA Bucharest, Pitești University Centre, in partnership with the National Polytechnic Institute of Toulouse, France

Oct. 2012–Feb. 2022 | Lecturer
University of Pitești, Faculty of Mechanics and Technology, Manufacturing and Industrial Management Department

- Taught Materials Science and Engineering, Manufacturing Technologies for Semi-Finished Products, Biomaterials, Composite Materials, Materials Characterisation, and Phase Transformations and Materials Microstructure; delivered laboratory classes and seminars, including in Materials Technology and Applied Thermodynamics in Metallurgy.
- Served on dissertation and doctoral committees; supervised undergraduate projects and dissertations; coordinated student research papers; directed the STM master's programme.

Oct. 2011–Feb. 2012 | Associate Lecturer
University of Pitești, Manufacturing and Industrial Management Department
Delivered practical classes in Materials Science and Engineering for the Transport Engineering programme.

Nov. 2002–Dec. 2007 | Full-time doctoral student; from Jan. 2007, Scientific Researcher, Grade II
University POLITEHNICA of Bucharest, Faculty of Applied Chemistry and Materials Science, Department of General Chemistry
Conducted General Chemistry laboratory classes for several faculties (Aerospace Engineering, Transport Engineering, Automation and Computer Science, Electrical Engineering, Mechanical Engineering, Biotechnical Systems Engineering, and Engineering and Management of Technological Systems); delivered seminars, contributed to department activities, and carried out research.

Oct. 2019–July 2021 | Trainer and tutor
ROSE project "I Will Be an Engineer! – FMT", University Grant Scheme SGU/NC/II, beneficiary: University of Pitești, agreement AG 179

Oct. 2020–July 2022 | Trainer and CCOC lead tutor
ROSE project "First Steps towards Industrial Engineering and Management", SGU/SS/III, beneficiary: University of Pitești, agreement AG 353

Oct. 2023–June 2024 | Trainer and CCOC lead tutor
ROSE project "First Steps towards Industrial Engineering and Management" (ING-PIM), beneficiary: POLITEHNICA Bucharest – Pitești University Centre, agreement AG399/SGU/SS/2023

May–June 2025 | Trainer for student remedial programmes
PRIM STUD project, MySMIS code 326436, Education and Employment Programme 2021–2027, focused on preventing early school leaving and increasing disadvantaged groups' access to quality tertiary education

EDUCATION AND TRAINING

2002–2006 | PhD in Materials Science and Engineering / Materials Engineering
Joint doctoral programme: University POLITEHNICA of Bucharest, Faculty of Applied Chemistry and Materials Science, and Institut National Polytechnique de Toulouse, France
Thesis: "From tri-isopropoxide to aluminium oxides by chemical vapour deposition: process, composition and properties of deposited coatings." Supervisors: Prof. Ioana Demetrescu and Prof. Alain Gleizes. Coursework and expertise: bioinformatics, applied electrochemistry, chemical equilibrium, spectroelectrochemical methods; MOCVD deposition of thin alumina films; characterisation using EDS, EMPA, ERDA-RBS, XRD, SEM, TEM, FTIR, TGA and AFM; corrosion testing and cyclic voltammetry. Complementary "Vers mon métier" training: project management and personal/professional project assessment (INP Toulouse). Four research placements at CIRIMAT, INP Toulouse: 8 months (Oct. 2003–May 2004), Romanian government scholarship; 6 months (Jan.–June 2005), funded by INP/CIRIMAT; 8 months (Sept. 2005–Apr. 2006), CNCSIS TD project no. 89, EGIDE/Brâncuși and CIRIMAT; 4 months (Sept.–Dec. 2006), CNCSIS TD project no. 89 and CIRIMAT.

2000–2002 | French-Romanian Master's Degree in Materials Science and Technology
University of Pitești, Faculty of Mechanics and Technology; in partnership with INP Toulouse
Four-month research placement (March–June 2002) with the Thin Film Materials team, ENSIACET/CIRIMAT, through the Socrates-Erasmus programme. Coursework: applied thermodynamics in metallurgy, materials characterisation, mechanical properties of surfaces, structural transformations, durability and corrosion, mechanical behaviour, protective coating technologies, and experimental research.

1996–2000 | Bachelor's Degree in Physical Chemistry
University of Pitești
Coursework: chemistry of metals and non-metals, physical chemistry of macromolecules, colloids and interfaces, instrumental analysis, biochemistry, electrochemistry, catalysis, chemical technology, thermodynamics, mechanics, molecular physics and heat, electricity and magnetism, optics, plasma physics, solid-state physics, atomic and nuclear physics, physical electronics, analytical monitoring of pollutants, programming, and pedagogy.

13–28 Feb. 2023 | Certificate of Completion – Entrepreneurial Skills
Ministry of Labour and Social Solidarity; Ministry of Education; Junior Achievement of Romania

17 Feb. 2025 | Specialist Course "Oracle Primavera P6, Basic & Advanced"
FIA Integra SRL, Bucharest

SKILLS

Languages | Romanian – native; French – C2; English – B1/B2
Self-assessment: French C2 in listening, reading, interaction, spoken production and writing; English B1 in listening and interaction, B2 in reading, spoken production and writing.

Social and organisational skills | Communication, collaboration and adaptability
Teamwork, responsibility, adaptability to multicultural environments and stressful situations, sociability, receptiveness, and creativity in conflict resolution; analysis, evaluation, improvement, decision-making and organisation, developed in part through volunteering.

Personal skills | Flexibility and commitment
Self-motivated learner; decision-making, monitoring, coordination, evaluation and communication; adaptability, commitment, teaching and interpersonal skills, analysis and synthesis.

SCIENTIFIC ACTIVITY

Jan. 2007–Aug. 2008 | Scientific Researcher, Grade II
CEEX – Postdoctoral Human Resources Programme

Micro- and nanostructures produced through chemical and electrochemical bioactivation, with applications in regenerative medicine. University POLITEHNICA of Bucharest, Faculty of Applied Chemistry and Materials Science, Department of General Chemistry.

Nov. 2002–Dec. 2006 | Joint doctoral researcher
University POLITEHNICA of Bucharest and Institut National Polytechnique de Toulouse, CIRIMAT–UMR-CNRS 5085, ENSIACET

Doctoral research on the production of aluminium oxides by chemical vapour deposition; supervisors: Prof. Ioana Demetrescu and Prof. Alain Gleizes.

2005–2006 | Principal Investigator – CNCSIS TD Grant, no. 89
University POLITEHNICA of Bucharest, Faculty of Applied Chemistry and Materials Science

“Aluminium oxide coatings by CVD for preparing supported catalysts used in the automotive industry to reduce pollution and ensure environmental safety.”

2010–2012 | Principal Investigator – PN II Postdoctoral Project PD_195/2010, no. 08
University of Pitești, Advanced Materials Research Centre (CCMA)

“Contributions to plasma electrolytic oxidation treatments for the development of biocompatible materials”; funding: RON 322,642.

2021–2022 | Principal Investigator – CIPCS 2021, acronym EVMFCS
University of Pitești, Faculty of Mechanics and Technology

“Development and validation of a model for building scientific research competencies among master’s students”; Social Sciences field; funding: RON 50,000.

RESEARCH OUTPUT AND PROJECTS

- 30 articles in Thomson Reuters ISI-indexed journals and ISI Proceedings volumes: 22 journal articles (8 as first author) and 8 papers in ISI Proceedings volumes (5 as first author).
- 21 articles in journals and proceedings indexed in other international databases; 19 conference posters (12 as first author).
- Teaching and specialist publications: 1 book, 6 textbooks/teaching materials, and 6 laboratory manuals.
- Grants and collaborations: CNCSIS TD grant no. 89 (2005–2006; RON 7,500 in 2005 and RON 9,000 in 2006); postdoctoral project PD_195/2010 (RON 322,642); CIPCS EVMFCS (RON 50,000).
- Member of the international PAI bilateral Brâncuși project 08619SL (2004–2006); member of 8 research project teams (2003–2010) and two contracts with the socio-economic sector (2013–2014).
- Short-term expert on the PERFORM project – “Sustainable Performance in Doctoral and Postdoctoral Research”, POSDRU/159/1.5/S/138963 (2014–2015).
- Member of the educational project FMT Engineer and Entrepreneur, Success for the Future! (2021), Engineering and Management field, RON 20,000.
- Member of the Horizon 2020 / Access SFERA III project, “The influence of sintering temperature on the structure and properties of ceramics”, SURPF2201300021, grant 823802 (2022–2023), RON 130,000.

March 2026